

YLITORNIO MELTOSJARVI, Finland

WMO: 028120

Lat: 66.529N

Lon: 24.650E

Elev: 305

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-19.6	-31.9	0.9	-25.5	-25.2	1.4	-19.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	79.1	62.6	75.1	60.9	71.5	59.0	65.6	73.8	63.3	71.7	61.2	68.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	86.3	68.6	60.3	79.1	65.1	58.4	73.7	63.2	30.6	73.7	28.8	72.1	27.4	68.3	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-29.9	82.8	6.5	4.9	-34.6	86.4	-38.4	89.3	-42.0	92.0	-46.7	95.6		
(5)			-30.2	68.5	6.4	2.8	-34.8	70.5	-38.5	72.2	-42.1	73.8	-46.7	75.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	8.5	12.1	20.8	32.6	45.3	53.7	59.7	54.9	47.0	33.5	24.0	16.8		
	DBStd	19.62	13.76	15.23	10.43	7.17	8.18	6.72	5.93	6.29	6.22	7.75	10.79	12.68		
	HDD50	6480	1288	1062	905	521	195	35	3	24	127	512	780	1028		
	HDD65	11273	1753	1482	1370	971	612	344	184	317	540	977	1230	1493		
	CDD50	714	0	0	0	0	48	147	303	178	38	0	0	0		
	CDD65	31	0	0	0	0	0	6	19	6	0	0	0	0		
	CDH74	451	0	0	0	0	37	85	276	53	0	0	0	0		
	CDH80	83	0	0	0	0	4	17	56	6	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	22.7	1.5	1.4	1.3	1.3	1.5	2.8	3.1	2.5	2.0	1.9	2.0	1.7		
(15) Precipitation	PrecMax	30.2	2.4	3.1	2.4	2.8	3.7	5.2	4.9	5.5	4.3	3.9	3.5	3.6		
	PrecMin	17.2	0.5	0.2	0.2	0.3	0.3	0.3	1.1	0.9	0.6	0.2	0.5	0.4		
	PrecStd	3.9	0.6	0.7	0.6	0.7	0.7	1.2	1.0	1.3	1.1	0.9	0.8	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	40.3	43.2	61.4	78.6	82.1	85.7	80.3	65.7	51.0	43.1	37.8		
		MCWB	33.2	37.0	36.6	46.9	56.8	63.8	66.7	67.7	56.8	47.6	42.2	36.8		
	2%	DB	32.0	35.6	39.5	51.8	72.9	75.8	80.9	74.4	62.6	48.9	40.2	34.7		
		MCWB	31.7	34.1	34.3	41.6	56.9	59.8	63.4	62.0	54.4	46.4	39.2	34.3		
	5%	DB	30.1	33.1	36.7	47.6	66.1	71.1	76.7	70.1	59.8	46.8	37.7	33.2		
		MCWB	29.7	32.5	32.9	39.6	52.9	56.8	61.9	59.5	54.5	45.1	37.4	32.9		
	10%	DB	27.2	31.3	34.2	43.7	60.6	66.9	73.0	66.2	57.1	44.3	35.9	32.2		
		MCWB	26.8	30.9	31.9	37.0	49.2	55.1	61.0	58.2	53.0	42.8	35.6	32.0		
	0.4%	WB	33.3	37.1	37.6	47.2	59.9	65.2	69.3	68.0	58.9	49.7	42.8	36.9		
		MCDB	33.8	39.8	42.2	59.8	73.4	76.8	78.6	76.5	61.8	49.9	42.9	37.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.7	34.1	35.2	42.6	57.1	62.3	66.5	64.1	56.9	46.8	39.4	34.3		
		MCDB	32.0	35.0	38.2	50.0	70.1	73.7	73.7	70.5	59.5	48.0	40.0	34.5		
	5%	WB	29.7	32.6	33.7	40.1	54.1	59.9	64.5	61.5	55.4	45.1	37.4	33.1		
		MCDB	30.1	33.2	35.5	46.7	64.8	67.2	73.8	66.8	58.4	46.7	37.6	33.2		
	10%	WB	26.8	30.8	32.5	37.8	50.8	57.1	62.5	59.7	53.7	42.9	35.6	32.0		
		MCDB	27.1	31.3	34.1	42.4	58.7	63.5	70.3	64.0	56.6	44.1	35.9	32.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	15.5	19.3	17.5	18.7	17.7	19.1	18.2	14.7	10.7	10.2	13.0		
		MCDBR	13.3	9.9	16.6	22.8	29.3	25.4	26.7	24.0	20.1	13.1	7.8	9.1		
		MCWBR	13.4	8.7	12.6	13.7	14.7	11.5	12.3	12.5	13.1	10.9	7.7	9.1		
		MCDBR	13.5	9.7	11.9	21.7	26.5	21.0	21.5	18.1	14.8	11.3	7.8	8.9		
	5% WB	MCWBR	13.6	8.6	9.5	13.4	13.7	10.7	11.5	10.6	10.3	9.9	7.7	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.199	0.267	0.298	0.333	0.343	0.346	0.359	0.342	0.317	0.290	0.345	0.360		
	taud		1.938	2.108	2.178	2.253	2.449	2.469	2.450	2.481	2.536	2.351	2.645	2.680		
	Ebn at Noon		64	181	238	261	272	274	266	259	240	178	68	17		
	Edh at Noon		5	17	27	32	30	30	30	26	19	14	5	3		
(44) All-Sky Solar Radiation	RadAvg		21	171	617	1140	1492	1650	1554	1128	595	210	42	2		
	RadStd		2	30	64	95	140	128	146	98	67	41	5	0		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page